# TECHNICAL DESIGN OPERATION & FORMULA WORKSHEET

## Technical Math & Architecture Schematic Design

**National Ice Ball League Sport, Inc. (dba: BIM-APES Software / NIBLS, Inc.)**

Alternate Names: National Intelligent Ball League Sports — National Intelligent Blockchain League Sports

**Licensor & Sole IP Owner: Dashawn Ramel Bledsoe**

## Executive Technical Overview

This worksheet establishes the quantitative specifications, signal calibration mathematics, valuation factor weights, dynamic clearing split rules, and black-box execution timelines for the **BIM-APES (Biometric Index Market — Athletic Performance Exchange Systems)** and **PhysioBioFin Operating Stack**. Built upon high-density 1000Hz biometric telemetry, real-time spatial computing, and programmatic clearing rails, this architecture transforms physical performance moments into collateralized financial assets under zero-dilution IP governance.

### Module 1: 1000Hz Telemetry Ingestion & Universal Normalization Scale

Raw multi-sensor telemetry (ECG/HRV, IMU kinematics, force output, recovery velocity) polling at 1000Hz across wearable mesh devices (Catapult Vector 8, WHOOP, Oura, Garmin, Tech+Ball™) is transformed onto a universal dimensionless scale:

$$\text{Normalized Score} = \left( \frac{\text{Raw Metric} - \text{Min}}{\text{Max} - \text{Min}} \times 110 \right) - 10 \tag{1}$$

- **Negative Momentum Zone ($-10$ to $0$)**: Captures autonomic fatigue collapse, joint stress overload, physiological tilt, and injury drag.
- **Standard Performance Zone ($0$ to $100$)**: Maps physical readiness, active performance baselines, and explosive surge thresholds.

**Signal Quality Gate (WD-CS-E / Domain 9)**: Before normalization, raw streams pass through the Wearable Drift Cyber-Stability Index (WD-CS-E) to filter sensor jitter, network latency, and camera occlusion:

$$\text{Signal Integrity Index (SNI)} = \frac{\text{Signal Power}}{\text{Noise Power} + \text{Sensor Drift}} \tag{2}$$

## Module 2: BLEI-E Master Composite Risk Engine & Factor Weights

The **Biometric-Linked Equity Index (BLEI-E)** synthesizes real-time physical telemetry, stadium economics, virality, and compliance signals into an institutional composite score (0 to 1,000):

$$\text{BLEI-E Master Score} = \sum_{k=1}^{n} (w_k \times F_{k,i}) + \epsilon_i \tag{3}$$

$$\text{BLEI-E Composite} = (0.40 \times \text{PMF-E}) + (0.25 \times \text{CSE-E}) + (0.25 \times \text{FBR-E}) + (0.10 \times \text{SEF-E}) \tag{4}$$

| Factor Symbol | Weight (%) | Functional Domain & Performance Indicators |
|---|---|---|
| **PMF-E** | **40.0%** | Physiological Market Factor: VO2 max, sprint speed, HRV, force output. |
| **CSE-E** | **25.0%** | Cortisol Stress Equity Factor: Autonomic fatigue & stress drag. |
| **FBR-E** | **25.0%** | Fan Base Revenue Factor: Stadium economics, merch, & ticket yield. |
| **SEF-E** | **10.0%** | Sponsor Equity Factor: NIL brand recall & ad-slot conversion lift. |
| **WD-CS-E** | *Oracle Filter* | Wearable Drift Filter: Signal quality & telemetry stability (Domain 9). |

## Module 3: Intrinsic Valuation (Model NIL) vs. Marketplace Price (Market NIL)

The valuation core continuously calculates fundamental physical output (**Model NIL**) against marketplace pricing (**Market NIL**) using Fama-MacBeth 2-pass regressions and GRS test statistics:

$$\text{Model NIL}_{\mathbf{i}} = \alpha_i + \beta_{\text{PMF}} \cdot \text{PMF-E}_i + \beta_{\text{CSE}} \cdot \text{CSE-E}_i + \beta_{\text{SEF}} \cdot \text{SEF-E}_i + \epsilon_i \tag{5}$$

$$\text{Dislocation Spread } (\boldsymbol{\Delta}) = \text{Model NIL} - \text{Market NIL} \tag{6}$$

| Market Regime | Spread Condition | CPM Pricing / Surcharge | Quant Desk Action |
|---|---|---|---|
| **Bull Regime** | Model NIL > Market NIL | $1,200–$2,000 (+40% Bull Surcharge) | Long Pair Trade / Call O tion Buy |
| **Sideways** | Model NIL = Market NIL | $600–$1,000 (Base CPM) | High-Frequency Scalp (¡10ms) |
| **Bear Regime** | Model NIL < Market NIL | $300–$550 (+25% Risk-Event Surcharge) | Put Option / Pay-Fixed N Swap |

## Module 4: Dynamic Programmatic Ad Pricing (CPI & CPM Mathematics)

In-stream customized AR overlays and broadcast tickers are priced dynamically frame-by-frame based on performance intensity and Glow Ring momentum multipliers (0.5 to 2.0):

$$\text{Cost-Per-Impression (CPI)} = \frac{\text{Market NIL} \times \text{Momentum Factor}}{\text{Total Impressions}} \tag{7}$$

$$\text{Cost-Per-Mille (CPM)} = \left( \frac{\text{Total Campaign Cost}}{\text{Total Impressions}} \right) \times 1,000 \tag{8}$$

$$\text{Bull Regime CPM} = \text{CPM}_{\text{Base}} \times (1 + 0.40) = \$1,200\text{–}\$2,000 \tag{9}$$

$$\text{Risk-Event CPM (Bear)} = \text{CPM}_{\text{Floor}} \times (1 + 0.25) = \$300\text{–}\$550 \tag{10}$$

## Module 5: The 60/20/10/10 In-Game Micro-Rail Revenue Clearing Engine

All gross in-game fan tipping, boost credit purchases, and QR flash-sale redemptions clear programmatically on-chain (Solana sub-second micro-settlement / Ethereum Reg D rails) under the **60/20/10/10 Split Rule**:

| Allocation Bucket | Split (%) | Institutional Target Account & Purpose |
|---|---|---|
| **Athlete NIL Wallet** | **60.0%** | Direct Creator Pool & Performance Bonus Wallet. |
| **Team / League Liquidity Pool** | **20.0%** | Organization Operations & Stadium Liquidity Pool. |
| **Sponsor Royalty Bucket** | **10.0%** | Self-Liquidating Margin Account (Brand Cash-Back). |
| **Platform Reserve & Governance** | **10.0%** | BIM-APES System Fee & Governance Reserve. |

## Module 6: AR Risk Composite (ARC) & Biometric Candlestick Doji Mechanics

Spatial joint tracking and Biomech Capsule dynamics map directly into the **AR Risk Composite (ARC)** early-warning score:

$$\text{ARC} = (w_1 \cdot \text{SSI}) + (w_2 \cdot \text{CIS}) \tag{11}$$

where SSI is the Spatial Stability Index (gait symmetry) and CIS is the Capsule Integrity Score (capsule wobble).

**Biometric Candlestick Mapping (OHLC)**:

- **Open**: Baseline physiological readiness score (CSI or PRI).
- **High**: Peak force output, G-force impact, or maximum joint stress load (Top Arc).
- **Low**: Maximum autonomic fatigue or recovery suppression (Bottom Arc red gradient).
- **Close**: Post-effort heart-rate recovery (HRR) efficiency (Center Line slope).

**Fatigue Doji Trigger Condition**: A narrow-body candlestick following high-stress bursts or velocity breakouts ($> 105$ mph exit velocity or $> 18$ mph sprint) signals physical exhaustion:

$$\text{Trigger Condition: } |\text{Close} - \text{Open}| \leq \delta \quad \text{AND} \quad (\text{High} - \text{Low}) \geq \gamma \tag{12}$$

When confirmed, ‘EventRuleEngine‘ fires an automated webhook in under **500 milliseconds**:

```
Trigger Webhook: POST /sponsor/activation/track
{
  "event_type": "BIOMETRIC_CANDLESTICK_DOJI_TRIGGER",
  "execution_latency_ms": 480,
  "metric_threshold": "EXIT_VELOCITY > 105_MPH",
  "rule_engines": ["EventRuleEngine", "SponsorRuleEngine"],
  "clearing_rule": "60/20/10/10_SPLIT",
  "custodian_sweep": "10PCT_SPONSOR_MARGIN_ACCOUNT"
}
```

## Module 7: Black-Box Rule Architecture & 24 Algorithmic Execution Methods

The software operates under five trade-secret black-box rule engines (**FICO® score architecture**), returning validated score outputs without exposing raw algorithms:

1. **EventRuleEngine**: Real-time performance & trigger windows.
2. **BiometricRuleEngine**: Biometric identity & 1000Hz physical benchmarks.
3. **NILRuleEngine**: Valuation logic & market-grade NIL score units.
4. **OverlayRuleEngine**: Contextual broadcast/OTT customized graphics.
5. **SponsorRuleEngine**: Brand alignment, CTA deployment & royalty sweeps.

**Quantitative Hedge Desk Execution Timeline**: Positions execute off an Exit Score (0–100) updated every 50–200 ms:

| Timeframe | Quantitative Desk Action & Automated Execution Method |
|---|---|
| **0 – 10 Seconds** | Methods 17 & 18 (Micro-Drift & Capsule Pulse Scalping) monetize sub-second flinches. |
| **10 – 60 Seconds** | Methods 11 & 12 (Spatial Drift & Capsule Behavior Exits) trim unbalanced positions. |
| **60 – 180 Seconds** | Peak effort triggers webhook activation in $< 500$ms for QR mobile push. |
| **Multi-Minute** | Method 1 (Biometric Adaptive Trailing Stop) dynamically tightens exit stops as fatigue builds. |

## Module 8: Telemetry Derivative Exchange (TDX) & Option Pricing Mechanics

The TDX clearing layer trades **Dynamic Ad-Collateralized Synthetic Units (DAC-Units)**:

$$\text{Parametric Downside Swap (PDS) Payout} = \text{Notional} \times \max\left(0, 1 - \frac{\text{Actual Recovery Index}}{\text{Baseline Threshold}}\right) \tag{13}$$

When interaction latency ($L_t$) surges or biometric exertion ($PFI$) crashes ($Z$-score $< -2.5\sigma$), the Circuit Breaker trips, freezing ad delivery to protect sponsor principal and converting remaining margin funds into yield-bearing cash equivalents.

**LEGAL DECLARATION & IP PROTECTION NOTICE**

All software architectures, mathematical formulas, black-box rule engines, Biometric Candlestick Doji algorithms, and Tech+Ball™ hardware designs are 100% sole personal intellectual property of **Dashawn Ramel Bledsoe / National Ice Ball League Sport, Inc. (dba: BIM-APES Software / NIBLS, Inc.)** under sole inventor governance. Protected as trade secrets and defensive prior art disclosures (IPCOM000271355D and IPCOM000254433).